



**BAULKHAM HILLS
HIGH
SCHOOL**

2020

**YEAR 11
YEARLY
EXAMINATIONS**

Mathematics Extension

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen.
- Board-approved calculators may be used
- A reference sheet is provided at the back of this paper
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for careless or badly arranged work

Total marks: 70

Section I – 10 marks (pages 2 – 4)

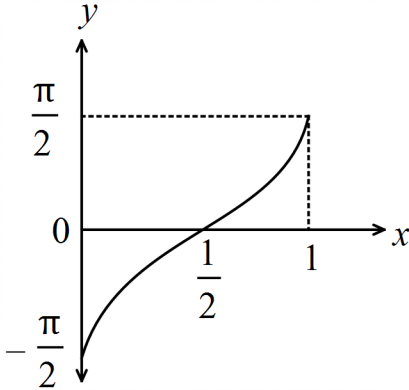
- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

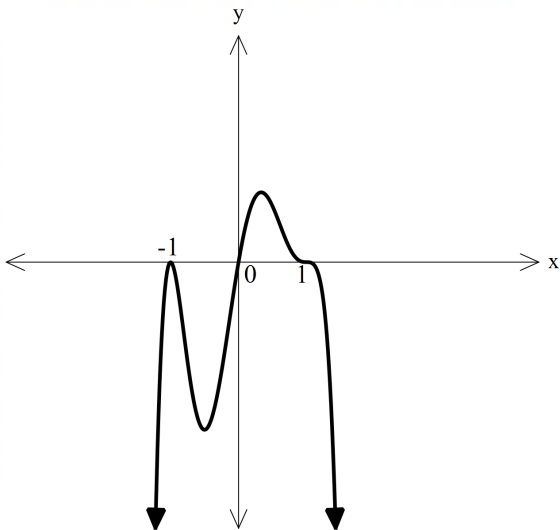
Section II – 60 marks (pages 5 – 8)

- Attempt Questions 11 – 14
 - Allow about 1 hour 45 minutes for this section
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Section I**10 marks****Allow about 15 minutes for this section.****Attempt all questions.****Answer each question on the multiple choice page of your answer booklet.**

1.	The polynomial $P(x) = x^4 - kx^2 - 2x + 33$ has $(x - 3)$ as a factor. What is the value of k ? (A) $-\frac{40}{3}$ (B) -12 (C) 12 (D) $\frac{40}{3}$	1
2.	The parametric equations $x = 3 + 4\cos\theta$, $y = 4\sin\theta - 4$ represent a circle, with centre C and radius r . Which of the following shows the correct centre and radius? (A) $C(-3,4)$ and $r = 4$ (B) $C(3,-4)$ and $r = 4$ (C) $C(-3,4)$ and $r = 16$ (D) $C(3,-4)$ and $r = 16$	1
3.	The number of worms W in a worm farm at time t days after the farm is set up is given by $W = 3000e^{0.05t}$. The rate of increase in the population after 6 hours is approximately: (A) 152 worms/hour (B) 202 worms/hour (C) 152 worms/day (D) 202 worms/day	1
4.	If $f(x) = \frac{3+e^{2x}}{4}$, which of the following is equal to $f^{-1}(x)$? (A) $\log_e(4x - 3)$ (B) $\frac{1}{2}\log_e 4x - 3$ (C) $\log_e \frac{4x-3}{2}$ (D) $\log_e \sqrt{4x - 3}$	1

5.	At a family gathering, 6 adults and 3 children are randomly seated around a circular table. What is the number of possible seating arrangements if none of the children are to sit together? (A) 720 (B) 2400 (C) 14400 (D) 86400	1
6.	The diagram shows the graph of a function $y = f(x)$.  Which function could the graph represent? (A) $y = -\cos^{-1}(2x - 1)$ (B) $y = \sin^{-1}(2x - 1)$ (C) $y = \sin^{-1}(x - 1)$ (D) $y = -\cos^{-1}(x - 1)$	1
7.	$2\cos 6\theta \sin 2\theta$ is equivalent to (A) $\cos 4\theta + \cos 8\theta$ (B) $\cos 8\theta - \cos 4\theta$ (C) $\sin 8\theta + \sin 4\theta$ (D) $\sin 8\theta - \sin 4\theta$	1

8.	Which of the following could be the equation of the graph given below?  (A) $y = -\frac{x}{2}(1-x)^3(1+x)^2$ (B) $y = -\frac{x}{2}(1-x)^2(1+x)^3$ (C) $y = 2x(1+x)^3(x-1)^2$ (D) $y = 2x(1-x)^3(x+1)^2$	1
9.	Which inequality has the same solution $x + 3 + x - 4 = 7$? (A) $2x - 1 \geq 7$ (B) $x^2 - x - 12 \leq 0$ (C) $\frac{7}{3-x} \geq 1$ (D) $\frac{1}{x-4} - \frac{1}{x+3} \leq 0$	1
10.	What is the coefficient of x^3 in the expansion of $(1 + x + x^2)^5$? (A) 10 (B) 20 (C) 30 (D) 40	1
End of Section I		

Section II

60 marks

Attempt questions 11-14

Allow about 1 hour 45 minutes for this section.

Answer each question on the appropriate page in the answer booklet. Extra writing paper is available.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 Marks)

Use the Question 11 section of the writing booklet.

- a) The parametric equations of a curve are $x = 4 - t$, $y = (1 - t)^2$. **2**
Find the Cartesian equation of the curve, simplifying your answer.
- b) Solve the inequalities
- (i) $\frac{2}{x-5} \geq 1$ **3**
- (ii) $\frac{7}{9-x^2} > -1$ **3**
- c) Simplify $\frac{\cos 2\alpha - \cos \alpha + 1}{\sin 2\alpha - \sin \alpha}$ **3**
- d) A group of 12 students is to be divided into discussion groups.
- (i) In how many ways can the discussion groups be formed if there are 7 people in one group and 5 people in the other? **1**
- (ii) In how many ways can the discussion groups be formed if there are 3 groups containing 4 people each? **2**
- (iii) In how many ways can a president and vice president be chosen for the group of 12? **1**

End of Question 11

Question 12 (15 marks)

Use the Question 12 section of the writing booklet.

a) One of the roots of the equation $x^3 + kx^2 + 1 = 0$ is the sum of the other two roots.

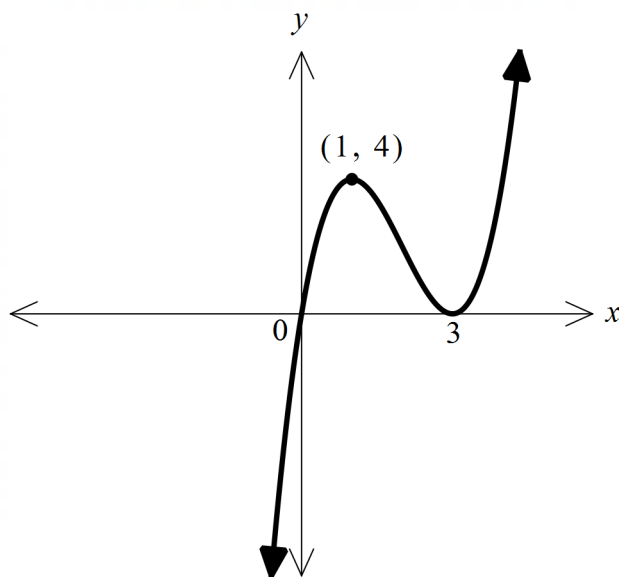
(i) Show that $x = -\frac{k}{2}$ is a root of the equation.

2

(ii) Find the value of k .

2

b) The diagram shows the graph of a function $y = f(x)$



(i) Draw the graph of $y = \frac{1}{f(x)}$ showing all important features.

3

(ii) Draw the graph of $y = f(|x|)$ showing all important features.

2

c) Find the coefficient of x^3 in the the expansion of $(3 - 4x)^7$

2

d) All the letters of the word DEFINITION are arranged in a row.

(i) How many distinct arrangements are possible?

1

(ii) In how many of these arrangements are all the vowels together?

2

(iii) What is the probability that the letter D appears somewhere to the left of the letter T?

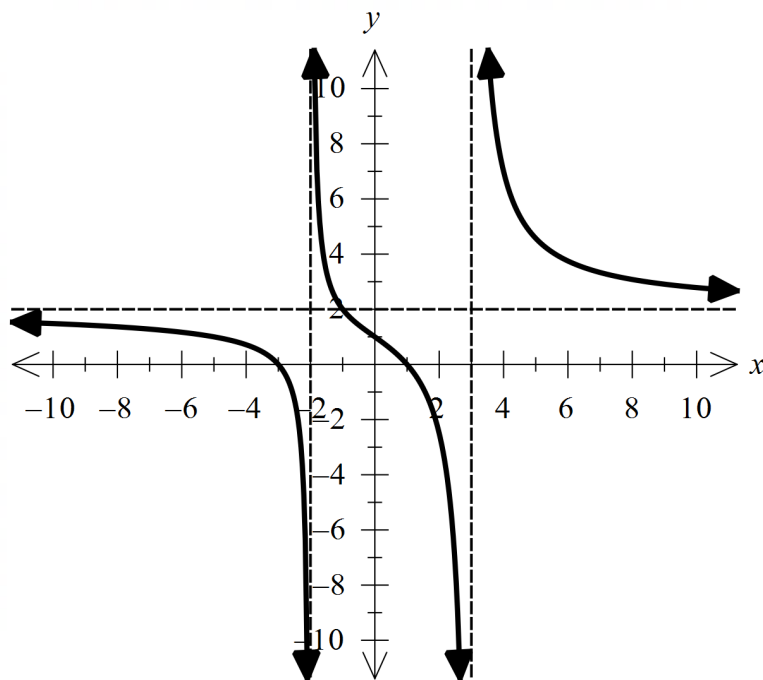
1

End of Question 12

Question 13 (15 Marks)

Use the Question 13 section of the writing booklet.

- a) The graph of $y = \frac{k(x-a)(x-b)}{(x-c)(x-d)}$ is shown, where k, a, b, c and d are constants.



- | | | |
|-------|--|---|
| (i) | State the values of a, b, c and d . | 2 |
| (ii) | Find the value of k . Justify your answer. | 1 |
| | | |
| b) | The equation $4x^3 - 27x + k = 0$ has a double root. Find the possible values of k . | 3 |
| | | |
| c) | If $(1 + kx)^n = 1 - 12x + 60x^2 - \dots$, find the values of k and n . | 3 |
| | | |
| d) | N is the number of animals in a certain population at time t years. The population satisfies the equation $\frac{dN}{dt} = -k(N - 1000)$, for some constant k . | |
| (i) | Verify that $N = 1000 + Ae^{-kt}$ (where A is constant) is a solution of the above equation. | 1 |
| (ii) | From an initial population of 2500 animals, after 2 years the population has fallen to 2200. Determine the values for A and k . | 2 |
| (iii) | Find when the population falls to 1200. Answer in years correct to 1 decimal place. | 2 |
| (iv) | Sketch the graph of population against time. | 1 |

End of Question 13

Question 14 (15 Marks)

Use the Question 14 section of the writing booklet.

- a) Express $\cot \frac{A}{2} - 2\cot A$ as a single trigonometric ratio. **2**
- b) If α, β and γ are the roots of $3x^3 + 4x^2 - 5x - 8 = 0$, find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$. **2**
- c) As a sphere of radius r is being heated, it expands such that its surface area S is increasing at a constant rate of $1.5 \text{ mm}^2\text{s}^{-1}$.
- (i) Show that $\frac{dr}{dt} = \frac{3}{16\pi r}$. **2**
- (ii) Hence find the rate at which its volume is increasing when its radius is 60mm. **2**
- d) Find the exact value of $\cos \left[\sin^{-1} \frac{5}{13} + \tan^{-1} \frac{3}{5} \right]$. **3**
- e) Consider the expression: **4**
- $$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \dots + \binom{n}{n-1}x^{n-1} + \binom{n}{n}x^n$$
- Multiplying by x , another identity is obtained:
- $$x(1+x)^n = \binom{n}{0}x + \binom{n}{1}x^2 + \binom{n}{2}x^3 + \binom{n}{3}x^4 + \dots + \binom{n}{n-1}x^n + \binom{n}{n}x^{n+1}$$
- Find the function $F(n)$ such that

$$2\binom{n}{1} + 3\binom{n}{2} + 4\binom{n}{3} + \dots + n\binom{n}{n-1} = (n+2)F(n)$$

End of Examination

SECTION I

1. $P(3) = 81 - 9k - 6 + 33 = 0$

$$108 - 9k = 0$$

$$k = 12$$

(C)

2. $\cos \theta = \frac{x-3}{4}$, $\sin \theta = \frac{y+4}{4}$

Since $\cos^2 \theta + \sin^2 \theta = 1$

$$\left(\frac{x-3}{4}\right)^2 + \left(\frac{y+4}{4}\right)^2 = 1$$

$$(x-3)^2 + (y+4)^2 = 16$$

$\therefore C(3, -4)$, $r = 4$ (B)

3. 6 hours $\Rightarrow t = 0.25$ days

$$\text{Rate} = \frac{dW}{dt} = 3000e^{0.05t} \times 0.05$$

$$= 150e^{0.05t}$$

$$= 150e^{0.05 \times 0.25} \quad \text{when } t = 0.25$$

$$\approx 152 \text{ worms/day}$$

$\leftarrow$ (since t was measured in days,

(C)

4. $f: y = \frac{3+e^{2x}}{4}$

$$f^{-1}: x = \frac{3+e^{2y}}{4}$$

$$4x-3 = e^{2y}$$

$$2y = \ln(4x-3)$$

$$y = \frac{1}{2} \ln(4x-3)$$

$$= \ln \sqrt{4x-3}$$

(D)

5. Arrange adults in circle $5!$

Place 3 children into the 6P_3 or $6 \times 5 \times 4$
6 spaces between adults

$$\therefore \text{No. of ways} = 5! \times {}^6P_3 = 14400 \quad \textcircled{C}$$

6. $\textcircled{B}$ (can be confirmed by substituting to find endpoints)

7. From ref. sheet $2 \cos A \sin B = \sin(A+B) - \sin(A-B)$

$$\therefore 2 \cos 6\theta \sin 2\theta = \sin 8\theta - \sin 4\theta$$

$\textcircled{D}$

8. Must have factors $(x+1)^2$, x and $(x-1)^3$

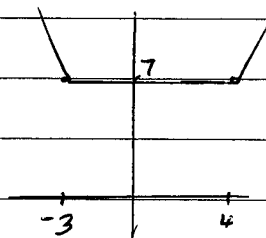
and a negative leading coefficient

$$\therefore y = kx(x+1)^2(x-1)^3 \quad \text{where } k < 0$$

$$\text{or } y = kx(x+1)^2(1-x)^3 \quad \text{where } k > 0$$

$\textcircled{D}$

9. $y = |x+3| + |x-4|$ sketch



$$|x+3| + |x-4| = 7$$

← when $-3 \leq x \leq 4$

Eliminate A, since its soln will be: $\leftarrow \bullet \bullet \rightarrow$

B is the correct soln



$\textcircled{B}$

C can be eliminated since $x \neq 3$ in its solution

D can be eliminated because $x \neq 4, -3$ in its solution

$$10. \quad ((1+x) + x^2)^5$$

$$= \binom{5}{0}(1+x)^5 + \binom{5}{1}(1+x)^4(x^2) + \binom{5}{2}(1+x)^3(x^2)^2$$

$$+ \binom{5}{3}(1+x)^2(x^2)^3 + \binom{5}{4}(1+x)(x^2)^4 + \binom{5}{5}(x^2)^5$$

$$= \binom{5}{0}(1+x)^5 + \binom{5}{1}(1+x)^4 x^2 + \binom{5}{2}(1+x)^3 x^4 + \dots$$

The only terms above that will include x^3 in their expansions are

$$\binom{5}{4}(1+x)^4 x^2 \quad \text{and} \quad \binom{5}{0}(1+x)^5$$

$$= 5x^2(1+x)^4 \quad \quad \quad = (1+x)^5$$

Terms containing x^3 will be

$$5x^2 \cdot \binom{4}{1}x + \binom{5}{3}x^3$$

$$= 20x^3 + 10x^3$$

$$= 30x^3$$

$\therefore$ Coefficient = 30 (C)

- | | |
|-----|------|
| 1 C | 6 B |
| 2 B | 7 D |
| 3 C | 8 D |
| 4 D | 9 B |
| 5 C | 10 C |

Question 11

a) $x = 4 - t \therefore t = 4 - x$

$$\begin{aligned} y &= (1-t)^2 \therefore y = (1 - (4-x))^2 \\ &= (-3+x)^2 \\ &= (x-3)^2 \\ &= x^2 - 6x + 9 \end{aligned}$$

(2) correct soln

(1) Substitute $t = 4 - x$

b) (i) $\frac{2}{x-5} \geq 1$

note:
 $x \neq 5$

$x(x-5)^2 :$

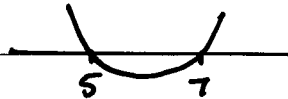
$$2(x-5) \geq (x-5)^2$$

$$2x - 10 \geq x^2 - 10x + 25$$

$$x^2 - 12x + 35 \leq 0$$

$$(x-5)(x-7) \leq 0$$

Roots: $x = 5, 7$



(3) correct soln

(2) • Solve quad. equation and attempt to find inequality

• $5 \leq x \leq 7$

(1) obtain quadratic

Soln: $5 < x \leq 7$

(ii) $\frac{7}{9-x^2} > -1$

note:
 $x \neq 3, -3$

$x(9-x^2)^2 :$

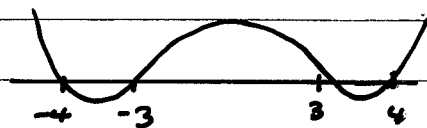
$$7(9-x^2) > -(9-x^2)^2$$

$$(9-x^2)^2 + 7(9-x^2) > 0$$

$$(9-x^2)(9-x^2+7) > 0$$

$$(3-x)(3+x)(4-x)(4+x) > 0$$

Roots $x = 3, -3, 4, -4$



(3) correct soln

(2) • correct roots

• solving a quartic eqn

(1) obtain a

quartic eqn and attempt to factorize

Soln: $x < -4, -3 < x < 3, x > 4$

$$c) \frac{\cos 2\alpha - \cos \alpha + 1}{\sin 2\alpha - \sin \alpha} = \frac{2\cos^2 \alpha - 1 - \cos \alpha + 1}{2\sin \alpha \cos \alpha - \sin \alpha}$$

$$= \frac{\cos \alpha (2\cos \alpha - 1)}{\sin \alpha (2\cos \alpha - 1)}$$

$$= \frac{\cos \alpha}{\sin \alpha}$$

$$= \cot \alpha.$$

(3) correct solution

(2) correct formulae for $\cos 2\alpha$ and $\sin 2\alpha$

• a correct formula for $\cos 2\alpha$ or $\sin 2\alpha$ and attempt to simplify

(1) correct formulae for $\cos 2\alpha$ or $\sin 2\alpha$

$$d) (i) {}^{12}C_7 \times {}^5C_5 \text{ (or just } {}^{12}C_7) = 792$$

(1) correct answer

$$(ii) \frac{{}^{12}C_4 \times {}^8C_4 \times {}^4C_4}{3!} = \frac{34650}{6}$$

(2) correct answer

$$= 5775$$

(1) correct numerator

$$(iii) 12 \times 11 = 132$$

(1) correct answer

Q12.

a) (i) Let roots = $\alpha, \beta, \alpha + \beta$.

$$\text{Sum of roots} = \alpha + \beta + (\alpha + \beta)$$

$$= 2\alpha + 2\beta$$

$$= 2(\alpha + \beta) = -k$$

$$\therefore \alpha + \beta = -\frac{k}{2}$$

Since $\alpha + \beta$ is one of the roots, $-\frac{k}{2}$ is one of the roots.

(2) correctly shown

(1) Attempts to use

sum of roots or

substitutes $x = -\frac{k}{2}$.

(ii) Product of roots $\alpha\beta(\alpha + \beta) = -1$

$$\alpha\beta\left(-\frac{k}{2}\right) = -1$$

$$\alpha\beta = \frac{2}{k}$$

Sum of pairs $\alpha\beta + \alpha(\alpha + \beta) + \beta(\alpha + \beta) = 0$

$$\frac{2}{k} + (\alpha + \beta)^2 = 0$$

$$\frac{2}{k} + \left(-\frac{k}{2}\right)^2 = 0$$

$$\frac{k^2}{4} = -\frac{2}{k}$$

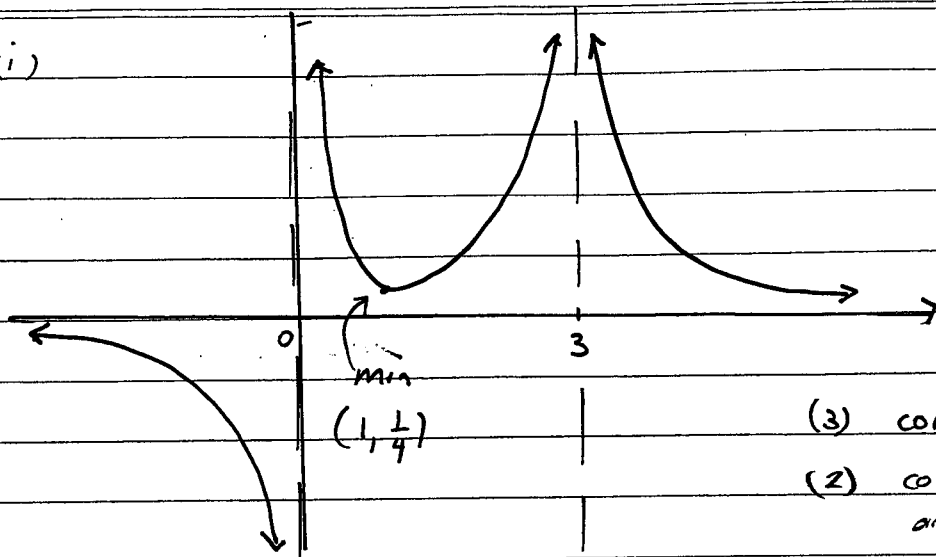
$$k^3 = -8$$

$$\underline{k = -2}$$

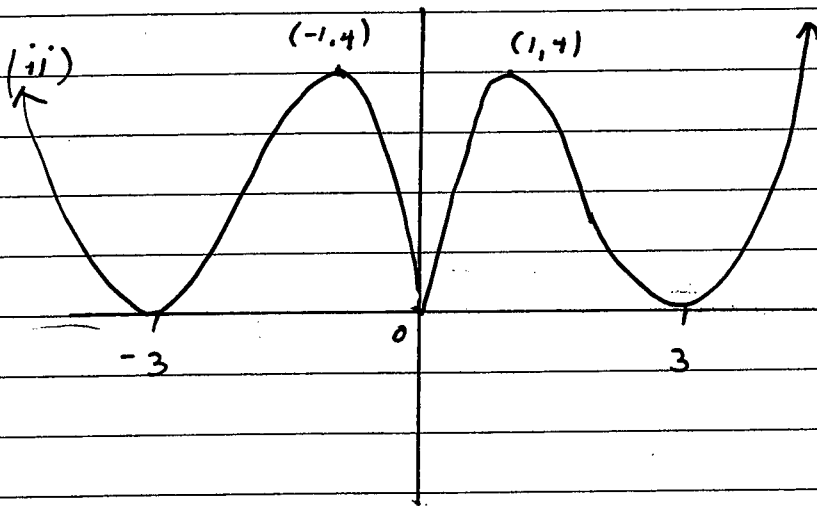
(2) correct soln

(1) attempts to use
sums + product.

b) (i)



- (3) correct graph (include turning point)
- (2) correct asymptotes and shape
- (1) correct asymptotes



- (1) correct graph, include (1, 4) and (-1, 4)

$$c) \left(\frac{7}{3}\right) \cdot 3^4 \cdot (-4x)^3 = 35 \times 81 \times -64x^3$$

$$\therefore \text{Coefficient} = -181440$$

- (2) correct soln
- (1) partially correct method.

d) DEFINTO

I N
I

10 letters

2 I, 2 N

5 vowels

5 consonants.

$$(i) \frac{10!}{3! 2!} = \underline{302400}$$

(1) correct answer

(ii) Arrange consonants

(D, F, N, T, N)

$$\frac{5!}{2!} = 60$$

^ ^ ^ ^ ^ ^

Place the vowels group in one of the 6 places indicated

6

Arrange the vowels within their group

$$\frac{5!}{3!}$$

$$\therefore 60 \times 6 \times \frac{5!}{3!} = 60 \times 6 \times 20$$

$$= \underline{7200}$$

(iii) The letter D can either appear to the left or to the right of the letter N, with equal probabilities.

$$\therefore \text{Probability} = \frac{1}{2}$$

(1) correct answer.

Question 13.

a) (i) $a = -3, b = 1$ (or vice versa)
 $c = -2, d = 3$ (or vice versa)

(2) All 4 correct

(1) One pair correct
 • Signs opp. but otherwise correct

(ii) $y = \frac{k(x-3)(x+1)}{(x-2)(x+3)}$

Sub. (0,1) :

$$1 = \frac{k \times -3 \times 1}{-2 \times 3}$$

$$-6 = -3k$$

$$k = 2$$

(1) correct soln + justification.

b) Let $P(x) = 4x^3 - 27x + k = 0$

If $P(x) = 0$ has a double root

then $P'(x) = 12x^2 - 27 = 0$ has a single root

} in common

$$P'(x) = 3(4x^2 - 9)$$

$$= 3(2x-3)(2x+3)$$

$$= 0 \text{ when } x = \frac{3}{2}, -\frac{3}{2}$$

If $x = \frac{3}{2}$,

then $4\left(\frac{27}{8}\right) - 27\left(\frac{3}{2}\right) + k = 0$

$$\therefore k = 27$$

If $x = -\frac{3}{2}$,

then $4\left(-\frac{27}{8}\right) - 27\left(-\frac{3}{2}\right) + k = 0$

$$k = -27$$

$\therefore$ The two possible values are $k = 27, -27$

(3) correct soln

(2) find $x = \pm \frac{3}{2}$ but not both solns given

(1) recognise common root exists and find $P'(x)$.

c) From T_2 : $\binom{n}{1}(kx)' = -12x$

$$nkx = -12x$$

$$nk = -12 \quad (1)$$

From T_3 : $\binom{n}{2}(kx)^2 = 60x^2$

$$\binom{n}{2}k^2 = 60$$

$$\frac{n(n-1)}{2} \cdot k^2 = 60$$

$$k^2 n(n-1) = 120 \quad (2)$$

From (1), $k = -\frac{12}{n}$

Into (2) $\left(-\frac{12}{n}\right)^2 \cdot n(n-1) = 120$

$$\frac{144}{n^2} \cdot n(n-1) = 120$$

$$144n^2 - 144n = 120n^2$$

$$24n^2 - 144n = 0$$

$$24n(n-6) = 0$$

$$n=0 \text{ or } n=6 \quad (n>0)$$

$\therefore$ Since $nk = -12$, $6k = -12$

$$k = -2$$

If $N = 1000 + Ae^{-kt}$, then:

d) (i) By differentiation, $\frac{dN}{dt} = \frac{d}{dt}(1000 + Ae^{-kt})$

$$= 0 + (Ae^{-kt}) \times (-k)$$

$$= -k(N-1000)$$

and $Ae^{-kt} = N-1000$

(1) correctly shown

(ii) $N = 1000 + Ae^{-kt}$

$t=0$

$N=2500$

$$2500 = 1000 + Ae^0$$

$$1500 = A$$

$$\therefore N = 1000 + 1500e^{-kt}$$

$$t = 2$$

$$N = 2200$$

$$2200 = 1000 + 1500e^{-2k}$$

$$0.8 = e^{-2k}$$

$$-2k = \ln 0.8$$

$$k = -\frac{1}{2} \ln 0.8 \doteq 0.1116$$

Store this value
for accuracy.

(2) correct A and k

(1) correct A or k

$$(iii) \quad N = 1000 + 1500e^{-0.1116 \dots t}$$

$$\text{If } N = 1200,$$

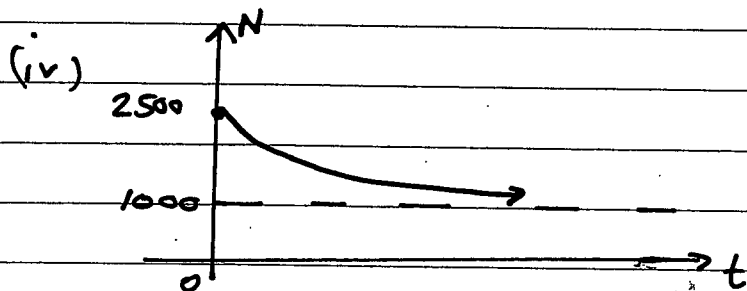
$$1200 = 1000 + 1500e^{-0.1116 \dots t}$$

$$\frac{2}{15} = e^{-0.1116 \dots t}$$

(2) correct soln

(1) obtain $e^{-kt} = \frac{2}{15}$

$$t = -\frac{1}{k} \ln \frac{2}{15} \doteq 18.1 \text{ years.}$$



(1) correct graph

with initial

value and asymptote

Question 14.

a) $\cot \frac{A}{2} = 2 \cot A$

$= \frac{1}{t} - 2 \left(\frac{1-t^2}{2t} \right)$ where $t = \tan \frac{A}{2}$

$= \frac{1 - (1-t^2)}{t}$

$= \frac{t^2}{t}$

$= t$

$= \tan \frac{A}{2}$

(2) correct soln

(1) attempt to use
 $\tan A = \frac{2t}{1-t^2}$

b) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$

$= \frac{-5/3}{8/3}$

$= -5/8$

(2) correct soln

(1) correct alg. fraction
+ correct product or
sum of pairs.

c) (i) $\frac{ds}{dt} = \frac{ds}{dr} \cdot \frac{dr}{dt}$ ($s = 4\pi r^2$, $\frac{ds}{dr} = 8\pi r$)

$1.5 = 8\pi r \times \frac{dr}{dt}$

$\frac{dr}{dt} = \frac{1.5}{8\pi r} = \frac{3}{16\pi r}$

(2) correctly shown

(1) forms a correct
eqn with derivatives

(ii) $\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$ ($V = \frac{4}{3}\pi r^3$, $\frac{dV}{dr} = 4\pi r^2$)

$\frac{dV}{dt} = 4\pi r^2 \times \frac{3}{16\pi r}$

$= \frac{3r}{4}$

$= \frac{3 \times 60}{4}$ when $r = 60$ mm

$= 45 \text{ mm}^3/\text{s}$

(2) correct soln

(1) correct eqn
for $\frac{dV}{dt}$ in terms
of r

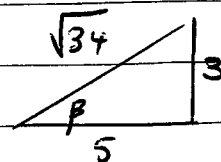
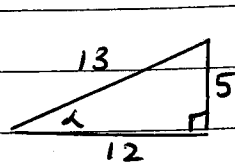
$$d) \cos \left(\underbrace{\sin^{-1} \frac{5}{13}}_{\alpha} + \underbrace{\tan^{-1} \frac{3}{5}}_{\beta} \right)$$

$$= \cos(\alpha + \beta)$$

$$= \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$= \frac{12}{13} \cdot \frac{5}{\sqrt{34}} - \frac{5}{13} \cdot \frac{3}{\sqrt{34}}$$

$$= \frac{45}{13\sqrt{34}}$$



(3) correct soln

(2) attempts to apply $\cos(\alpha \pm \beta)$ with values from Δs

(1) Δs with sides as shown.

e) Differentiating both sides of the 2nd identity:

$$LHS = x \cdot n(1+x)^{n-1} + (1+x)^n \cdot 1$$

$$= (1+x)^{n-1} (nx + 1+x)$$

$$RHS = 1 + 2\binom{n}{1}x + 3\binom{n}{2}x^2 + 4\binom{n}{3}x^3 + \dots + n\binom{n}{n-1}x^{n-1}$$

$$+ (n+1)\binom{n}{n}x^n$$

Substitute $x=1$ into both sides:

$$(1+1)^{n-1} (n+1+1) = \underline{1} + 2\binom{n}{1} + 3\binom{n}{2} + 4\binom{n}{3} + \dots + n\binom{n}{n-1}$$

$$2^{n-1} (n+2) = 2\binom{n}{1} + 3\binom{n}{2} + 4\binom{n}{3} + \dots + n\binom{n}{n-1} + \frac{(n+1) \cdot 1}{1}$$

$$\therefore 2\binom{n}{1} + 3\binom{n}{2} + 4\binom{n}{3} + \dots + n\binom{n}{n-1} + n+2$$

$$= 2^{n-1} (n+2) - (n+2)$$

$$= (n+2)(2^{n-1} - 1) \quad \therefore F(x) = 2^{n-1} - 1$$

(4) correct soln

(3) correctly substitutes $x=1$ into 2 correct expressions

(2) Correctly differentiates both sides

• Sub $x=1$, but only one side correct

(1) Correctly differentiates one side